Econometrics II — Problem Set 7

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Question 1. **Use `dataHW7_Problem1.mat` to complete this exercise. The data is identical to `dataHW6_Problem2.mat` — meaning the structure of the problem is the same as are the underlying parameters. The purpose of this problem is to estimate a simple dynamic discrete choice problem using CCPs. The data contain information on the choices (Y's), characteristics that do not vary over time (X_1's), and characteristics that do vary over time (X_{1t}'s). The context of the problem is a dynamic model of labor supply. Y takes values $\{0, 1, 2, 3\}$ reflecting retirement, flex-time, part-time, or full-time participation. The first column of X_1 allows for choice-specific constants, while the second column is an indicator for gender. X_{1t} is a time-varying measure of health. Both sets of X's are individual-level characteristics, so the coefficients on these variables must vary by choice. There is a cost to switching labor supply that is common across all switches. The choice in period 0 is given by LY. Once an individual retires (chooses option 0) they make no further choices: if $Y_{it} = 0$ then $Y_{it+1} = 0$. The columns of Y and X_{1t} indicate observations over time, ten periods in all, so that period 10 is a static problem. Individuals are uncertain about future health. Given X_{1t} and Y_t, the health transition is:**

$$X_{1,t+1} = \alpha_0 + \alpha_1 X_{1t} + \alpha_2 \mathbf{1}(Y_t = 2) + \alpha_3 \mathbf{1}(Y_t = 3) + \varepsilon_{t+1} \tag{1}$$

for $Y_t > 0$, where $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma)$ is unknown to the individual until time $t + 1$.

(a) Estimate the transition parameterts, the α's and σ.

Solution.

Table 1: OLS Estimates

	Parameter
α_0	0.015
α_1	-0.754
α_{pt}	0.035
α_{ft}	-0.257
σ	1.003

(b) Estimate the CCP's via a single logit model of retirement based on LY, X1, X1t, and a set of time dummies.

Solution.

Table 2: Part B: Conditional Choice Probability Logit Estimates

Variable	Coef.
X_1 (Constant)	-3.6991
X_2 (Gender)	-0.8556
X_t (Health)	-0.3494
Y_0 (Initial state)	0.0115
$\mathbf{1}(\bar{Y}_{t-1} = 2)$ (Part-time)	-0.1116
$\mathbf{1}(\bar{Y}_{t-1} = 3)$ (Full-time)	-0.0807
$t = 2$	-0.0443
$t = 3$	0.1227
$t = 4$	0.2846
$t = 5$	0.6496
$t = 6$	1.0620
$t = 7$	1.5239
$t = 8$	2.1992
$t = 9$	3.6510
Observations	
39,413	
Pseudo R^2	
0.2354	

(c) Use the CCP estimates and the future value terms calculated in (C) to estimate choice specific parameters, switching costs, and the discount factor via multinomial logit.

Solution.

Table 3: Part D: Dynamic Discrete Choice Estimates (CCP Method)

Parameter	Estimate	Truth
Flex-time ($j = 1$)		
Constant	-0.8940	-0.75
Gender	0.5936	0.25
Health	0.1695	0.25
Part-time ($j = 2$)		
Constant	-0.3305	-0.25
Gender	0.8890	0.50
Health	0.4162	0.50
Full-time ($j = 3$)		
Constant	-1.0477	-0.85
Gender	1.1255	0.75
Health	0.6673	0.75
Switching cost	0.4239	0.40
Discount factor β	0.7796	0.80